

STABLE, HIGH-ORDER EXPONENTIAL IMEX LINEAR MULTISTEP METHODS

SANAA MOUZAHER* AND PIERRE F. J. LERMUSIAUX†

Abstract. We develop and analyze implicit–explicit linear multistep methods for semilinear evolution equations with a prescribed linear–nonlinear decomposition. The linear dynamics are advanced by an integrating factor, whereas the nonlinear term is discretized explicitly, and a moment representation of the resulting multistep formula yields its complete local truncation-error expansion. On this basis, we construct an additive correction that cancels every error term through a prescribed order q .

The correction is distributed between the current and new time levels through a weight θ , with the new-level contribution treated linearly implicitly. This formulation produces a rational modewise amplification factor and enlarges the stability region relative to the explicit formula, while requiring only one linear solve with a frozen operator that is diagonal in the eigenbasis of the linear part.

Finite differences recover the nonlinear time derivatives from stored solution and nonlinear histories, while surrogate operators approximate the remaining terms in the derivative ladder. We derive the corresponding consistency condition, prove that a history depth $S \geq q$ preserves order q for every θ , and quantify the contribution of the surrogate error to the corrected local truncation error through the second finite-difference moment.

Companion-polynomial analysis provides amplification factors and modewise stability conditions for the base, exactly corrected, and surrogate-corrected schemes; it also identifies the parasitic roots introduced by the history reconstruction and explains their suppression through implicit weighting. For quadratic nonlinearities, the resulting formulas reduce to explicit polynomial symbols. Numerical experiments on the Burgers and Korteweg–de Vries equations and the Lorenz-63 system confirm the predicted convergence orders, enlarged stability regions, and computational advantages relative to fourth-order Runge–Kutta and exponential integrators.

Key words. implicit–explicit methods, linear multistep methods, integrating factors, linearly implicit corrections, stability analysis, local truncation error, data-driven numerical methods

MSC codes. 65L06, 65M12, 65M20, 68T07

1. Introduction. Spatial discretization of many time-dependent partial differential equations produces large systems of ordinary differential equations in which a stiff linear operator is coupled to an expensive nonlinear component. This structure motivates implicit–explicit (IMEX) discretizations, which treat the stiff contribution stably while advancing the nonlinear contribution explicitly.

Ascher, Ruuth, and Wetton systematize IMEX linear multistep methods (LMMs) in [4], building on the leapfrog–Crank–Nicolson and Adams–Bashforth–Crank–Nicolson combinations used in meteorology and Navier–Stokes computations [20, 21]. Subsequent analyses address stability [22] and monotonicity [23], while IMEX Runge–Kutta counterparts are developed in [24, 25, 26] and their order reduction is examined in [27].

Dahlquist establishes the order and zero-stability theories of multistep methods [1, 12], with standard presentations given in [2, 9, 3, 28]; the Adams and backward-differentiation families originate in [29, 30, 31].

When the linear operator is time independent and its exponential is efficient to apply, exponential integrators resolve the linear flow exactly and discretize only

*Multidisciplinary Simulation, Estimation, and Assimilation Systems (MSEAS) Group, Center of Computational Science and Engineering, Massachusetts Institute of Technology, Cambridge, MA 02139, USA (sanaamz@mit.edu).

†Multidisciplinary Simulation, Estimation, and Assimilation Systems (MSEAS) Group, Center of Computational Science and Engineering, Massachusetts Institute of Technology, Cambridge, MA 02139, USA (pierrel@mit.edu). Corresponding author.

the nonlinear term [6, 32]. The integrating-factor, or Lawson, formulation applies a classical method to the transformed variable [5], whereas exponential multistep methods [33, 34, 35, 36], exponential Runge–Kutta methods [14, 37], and exponential time differencing [38, 7, 8] directly integrate the variation-of-constants formula with φ -functions. Krylov techniques extend these methods to large nonnormal operators [39, 40]. This distinction is consequential for dispersive problems because the error of the Lawson formulation is independent of the linear multiplier, whereas exponential Runge–Kutta methods can exhibit order reduction [14].

A multistep discretization is efficient because it reuses stored solution values and nonlinear evaluations; however, classical order elevation requires a longer history and generally reduces the stability region [2]. We instead use the available history to correct the local truncation error of a low-order exponential multistep formula to a prescribed order without altering its base discretization.

The correction contains total time derivatives of the nonlinear term that are not directly available during time integration and, at high order, form intricate compositions of state and Fréchet derivatives through the multivariate Faà di Bruno formula [17, 18]. Because their exact evaluation by automatic differentiation requires a number of directional derivatives that grows quadratically with the order [19], we reconstruct the accessible derivatives from stored histories by finite differences and use a learned operator only for the highest unresolved derivative.

The proposed method belongs to the class of machine-learning-augmented schemes, in which a learned component corrects a coarse or low-order discretization without replacing the underlying solver [41, 42, 43, 44]. It is also related to learned closure models for dynamical systems and time integration [45, 46, 47]. In contrast to neural differential equations [48], the learned component enters only through a derivative isolated by the analysis, which allows consistency and stability to be established in terms of the approximation error before an architecture is selected. The supplementary materials consider translation-equivariant polynomial and linear convolution surrogates whose calibration reduces to a linear least-squares problem in the Volterra–Wiener sense [49, 50].

The second ingredient concerns the temporal placement of the correction. A current-level correction produces an explicit polynomial map with a bounded stability region, while reconstruction from stored levels introduces parasitic roots governed by Dahlquist’s root condition [1]. We therefore expand the same defect about the new level, as in Adams–Moulton correctors [30, 2], and combine the two expansions so that the resulting rational amplification factor has an unbounded stability region when the derivatives are exact.

The scheme realizes the new-level contribution linearly implicitly in the Rosenbrock sense [13, 9], using a frozen operator that is diagonal in the eigenbasis of the linear part. In the spirit of alternating-direction splittings [10, 51], this construction requires one diagonal solve per step but does not require the Jacobian of the nonlinearity.

1.1. Principal contributions. The principal contributions are the following.

1. We derive a moment representation of the local truncation error for explicit linear multistep discretizations of the integrating-factor equation, in which the defect at each order is a scalar coefficient multiplying a transformed nonlinear derivative centered at either the current or the new time level.
2. We combine the two representations through a weight θ . The resulting additive correction removes every multistep error term through a prescribed order

- q , with its new-level component realized linearly implicitly. The additional work consists of one nonlinear evaluation at a predicted level and one linear solve with a frozen operator diagonal in the eigenbasis of L .
3. We reconstruct the derivative ladder from S stored levels and one surrogate operator, prove that $S \geq q$ preserves order q for every θ , and derive the leading surrogate-induced error through the second finite-difference moment.
 4. We derive companion polynomials for the base, exactly corrected, and surrogate-corrected schemes, thereby obtaining the rational modewise amplification factor, an admissible surrogate-error tolerance, and the parasitic roots produced by the reconstruction, together with the weights that control these roots.

1.2. Organization. Section 2 introduces the semidiscrete problem, derivative identities, and exact exponential flow map. Section 3 develops the corrected exponential linear multistep methods and analyzes their order, local truncation error, and stability. Section 4 reconstructs the derivative ladder from stored histories and derives the associated error estimates and parasitic roots. Section 5 presents the numerical experiments. Section 6 summarizes the results.

2. Problem formulation and preliminary identities.

2.1. Semidiscrete problem and notation. We consider an autonomous nonlinear partial differential equation whose spatial discretization yields the finite-dimensional system

$$(2.1) \quad \partial_t u = \mathcal{F}(u) \quad \longrightarrow \quad \dot{u} = G(u) = Lu + N(u), \quad u(t) \in \mathbb{R}^M.$$

Throughout the analysis, the split (L, N) is prescribed [4], with L linear and time independent so that it can be treated exactly by an integrating factor [6]. We define the matrix exponential

$$(2.2) \quad E := e^{\Delta T L}$$

and assume that its integer powers E^j are available, as occurs when L is diagonal or arises from a spectral discretization. The nonlinear term N , which is treated explicitly and includes any static forcing, is assumed to be C^{S+1} near the trajectory, where S denotes the stored-history depth; constants may depend on the discretization size M . The general development imposes no polynomial, quadratic, or locality assumption on N , although the quadratic class $N(u) = P(u, u) + F$ is considered subsequently.

The surrogate stage of Sections 4–4.3 uses only the stored values $u(t_{n-i})$ and $N(u(t_{n-i}))$, whereas every factor E^j is subsequently applied by the time-integration scheme; see Sections 3, 3.4, and 4.3.

We write t_n for the current time and ΔT for the uniform time step, while parenthesized superscripts denote total time derivatives evaluated at t_n and primes denote Fréchet derivatives with respect to the state. The integers s , q , and S represent the multistep depth, target order, and stored-history depth, respectively. A tilde indicates transformation by the linear flow, a plain hat denotes a reconstructed or surrogate-dependent quantity, and modal symbols are reserved for the stability analysis. Unless stated otherwise, an approximation error is defined as the approximate quantity minus its exact counterpart, with all remaining notation introduced at first use.

2.2. Derivative identities. Because L is constant, differentiating (2.1) gives

$$(2.3) \quad u^{(k)} = Lu^{(k-1)} + N^{(k-1)}.$$

and unrolling this recursion to $u^{(0)} = u$ yields

$$(2.4) \quad \boxed{u^{(k)} = L^k u + \sum_{m=0}^{k-1} L^{k-1-m} N^{(m)}}.$$

We denote by $N^{(r)}(u)$ the symmetric r -linear Fréchet derivative and by $N^{(m)}$ the m th total time derivative of $N(u(t))$ at t_n ; the argument distinguishes the two uses. With $J(t) := N'(u(t))$ and $J^{(j)} := \frac{d^j}{dt^j} N'(u(t))|_{t_n}$, the Leibniz rule applied to $N^{(1)}(t) = J(t)u^{(1)}(t)$ gives the operator ladder

$$(2.5) \quad \boxed{N^{(k+1)} = \sum_{j=0}^k \binom{k}{j} J^{(j)} u^{(k+1-j)}}, \quad b_{k+1} := \sum_{j=1}^k \binom{k}{j} J^{(j)} u^{(k+1-j)},$$

where b_{k+1} collects the terms involving only $u^{(1)}, \dots, u^{(k)}$. The multivariate Faà di Bruno formula expresses both $N^{(m)}$ and $J^{(j)}$ as sums over set partitions of products of Fréchet derivatives and lower rungs [17, 18]; for example, $J^{(1)} = N''(u)[u^{(1)}, \cdot]$. Because evaluation of the ladder to order S requires $S(S+1)/2$ directional derivative actions [19], this cost motivates the reconstruction introduced in Section 4.

2.3. Exponential integration and exact flow map.

$$(2.6) \quad w(t) := e^{-(t-t_n)L} u(t), \quad \tilde{N}(t) := e^{-(t-t_n)L} N(u(t)).$$

Application of the product rule and (2.1) yields

$$(2.7) \quad \dot{w} = -L e^{-(t-t_n)L} u + e^{-(t-t_n)L} (Lu + N(u)) = e^{-(t-t_n)L} N(u) = \tilde{N}(t).$$

Thus, the transformed equation contains no explicit linear term, and at $t = t_n$ one has $w(t_n) = u(t_n)$ and $\tilde{N}(t_n) = N(u(t_n))$. The corresponding derivatives satisfy

$$(2.8) \quad w^{(k)} := \left. \frac{d^k w}{dt^k} \right|_{t_n} = \tilde{N}^{(k-1)}, \quad k \geq 1; \quad \tilde{N}^{(m)} := \left. \frac{d^m \tilde{N}}{dt^m} \right|_{t_n}.$$

The Leibniz rule on (2.6), with $\frac{d^j}{dt^j} e^{-(t-t_n)L}|_{t_n} = (-L)^j$, gives

$$(2.9) \quad \boxed{\tilde{N}^{(m)} = \sum_{j=0}^m \binom{m}{j} (-L)^j N^{(m-j)}}.$$

Since $N(u(t)) = e^{(t-t_n)L} \tilde{N}(t)$, the same argument with $+L$ gives the inverse

$$(2.10) \quad \boxed{N^{(m)} = \sum_{j=0}^m \binom{m}{j} L^j \tilde{N}^{(m-j)}}.$$

Relations (2.9) and (2.10) are mutually inverse by binomial inversion; consequently, $\tilde{N}^{(m)}$ involves only $N^{(0)}, \dots, N^{(m)}$ and the powers $L^0, \dots, L^m$, with no term of the form $L^k u$.

If

$$(2.11) \quad N(u) = P(u, u) + F,$$

with P symmetric bilinear and F constant, then $N'(u) = 2P(u, \cdot)$, $N'' = 2P$, $N''' \equiv 0$, and the ladders reduce to the Leibniz sums

$$(2.12) \quad N^{(m)} = \sum_{r=0}^m \binom{m}{r} P(u^{(r)}, u^{(m-r)}) \quad (m \geq 1), \quad J^{(j)} = 2P(u^{(j)}, \cdot) \quad (j \geq 0);$$

Because each operator action requires one bilinear evaluation, the ladder to order S requires $\sum_{m=1}^S (m+1)$ evaluations of P , each with the cost of one evaluation of N .

3. Higher-order exponential linear multistep methods.

3.1. Base exponential multistep method. With $t_{n+j} = t_n + j\Delta T$, (2.6) at the grid points gives

$$(3.1) \quad w^{n+j} := w(t_{n+j}) = e^{-j\Delta T L} u^{n+j} = E^{-j} u^{n+j}, \quad \tilde{N}^{n+j} := \tilde{N}(t_{n+j}) = E^{-j} N(u^{n+j}), \quad j \leq 1.$$

For $j \leq 0$, the exponent $-j = |j|$ is nonnegative, so $E^{|j|}$ transports the stored history forward to t_n and reduces to the identity when $j = 0$.

Applying an explicit s -step LMM [2, 6] to (2.7) gives

$$(3.2) \quad \sum_{j=-(s-1)}^1 a_j w^{n+j} = \Delta T \sum_{j=-(s-1)}^0 c_j \tilde{N}^{n+j}.$$

We adopt the normalization $a_1 = 1$ and require no separate family of implicit coefficients because the transformation (2.7) eliminates the linear term. Substituting (3.1) and then multiplying by E gives

$$(3.3) \quad \boxed{u^{n+1} = E \left[\sum_{j \leq 0} (-a_j) E^{-j} u^{n+j} + \Delta T \sum_{j \leq 0} c_j E^{-j} N(u^{n+j}) \right].}$$

Each time step therefore requires one new evaluation of N , at most s applications of E^{-j} to stored fields, and one application of E , but no linear-system solve.

The classical IMEX multistep methods of [4] treat L with an implicit formula having coefficients b_j and treat N with the explicit pair (a_j, c_j) . Because transformation (2.7) removes the implicit part, the integrating-factor counterpart is determined by the explicit pair alone; consequently, IMEX methods that differ only in b_j , such as CNAB and MCNAB, have the same integrating-factor counterpart.

Ascher, Ruuth, and Wetton parameterize the second-order two-step family by $\gamma \geq 0$ [4]:

$$(3.4) \quad a_1 = \gamma + \frac{1}{2}, \quad a_0 = -2\gamma, \quad a_{-1} = \gamma - \frac{1}{2}, \quad c_0 = \gamma + 1, \quad c_{-1} = -\gamma,$$

We normalize these coefficients by a_1 . Table 3.1 summarizes the instances used below.

TABLE 3.1
Base integrating-factor schemes used in the analysis and experiments.

Scheme	$p_\star$	γ	Description
IF-Euler	1	—	One-step Euler formula
IF-AB2	2	$\frac{1}{2}$	CNAB / Adams–Bashforth 2 pair
IF-SBDF2	2	1	Extrapolated Gear pair
IF-ARW _{3/2}	2	$\frac{3}{2}$	ARW family member

For IF-SBDF2, the normalized coefficients are $a_j = (1, -\frac{4}{3}, \frac{1}{3})$ and $c_j = (\frac{4}{3}, -\frac{2}{3})$.

3.2. Order conditions and local truncation error. Integrating (2.7) over one step gives

$$(3.5) \quad w(t_{n+1}) = w(t_n) + \int_0^{\Delta T} \tilde{N}(t_n + \xi) d\xi.$$

Expanding $\tilde{N}$ along the exact trajectory as $\tilde{N}(t_n + \xi) = \sum_{m \geq 0} \frac{\xi^m}{m!} \tilde{N}^{(m)}$ and integrating termwise yields

$$(3.6) \quad w(t_{n+1}) = w(t_n) + \sum_{m \geq 0} \frac{\Delta T^{m+1}}{(m+1)!} \tilde{N}^{(m)} = \sum_{p \geq 0} \frac{\Delta T^p}{p!} w^{(p)},$$

which, by (2.8), is the Taylor series of w ; multiplication by E then recovers the exact exponential flow map in the original variable.

For a smooth function ψ , we write $\sum_j a_j \psi(t_{n+j}) = \sum_{p \geq 0} \Delta T^p A_p \psi^{(p)}(t_n)$, where

$$(3.7) \quad A_p := \sum_j a_j \frac{j^p}{p!}, \quad C_p := \sum_j c_j \frac{j^p}{p!}.$$

These quantities are the Taylor coefficients of the generating polynomials at the shift operator [1, 2, 3]. Because the expansion is centered at t_n , this choice affects the error constants beyond the order of the method.

Inserting the exact solution into (3.2), with $C_{-1} := 0$ and (2.8), gives the residual (defect)

$$(3.8) \quad \rho^w := \sum_j a_j w(t_{n+j}) - \Delta T \sum_{j \leq 0} c_j \tilde{N}(t_{n+j}) = \sum_{p \geq 0} \Delta T^p \rho_p^w, \quad \rho_0^w = A_0 w,$$

$$\rho_p^w = \gamma_p \tilde{N}^{(p-1)}, \quad \gamma_p := A_p - C_{p-1}, \quad p \geq 1.$$

The method has order p_* if and only if

$$(3.9) \quad A_0 = 0, \quad \gamma_p = 0, \quad 1 \leq p \leq p_*.$$

Consequently, the scalar γ_p , which depends only on the coefficient families (a, c) , completely determines the defect at order p .

Assuming that the history is exact for $j \leq 0$, let w^{n+1} denote the value computed by (3.2), set $u^{n+1} = Ew^{n+1}$, and define the local errors by $\tau^w := w(t_{n+1}) - w^{n+1}$ and $\tau := u(t_{n+1}) - u^{n+1}$. Subtracting (3.2) from the defect identity cancels all history terms and, because c_{+1} is absent and $a_1 = 1$, yields $\tau^w = \rho^w$ together with

$$(3.10) \quad \tau^w = \sum_{p > p_*} \Delta T^p \tau_p^w, \quad \tau_p^w = \gamma_p \tilde{N}^{(p-1)},$$

and, by (3.1) at $j = 1$,

$$(3.11) \quad \tau = E \tau^w = \sum_{p > p_*} \Delta T^p E \gamma_p \tilde{N}^{(p-1)}.$$

This representation requires neither a recursive resolvent expansion nor the Jacobian of N and contains no term of the form $L^p u$ at any order; instead, L enters only through the binomial transformation (2.9) in $\tilde{N}^{(p-1)}$ and through the transport operator E .

For the instances listed above, (3.9) recovers $p_* = 1$ for IF-Euler and $p_* = 2$ for the family (3.4); the linear flow does not impose an additional order restriction because it is represented exactly by (3.3).

Expansion centered at the new level. Although the defect (3.8) is a single quantity, its expansion about t_{n+1} provides an equivalent representation in terms of new-level derivatives. With $t_{n+j} - t_{n+1} = (j-1)\Delta T$, define

$$(3.12) \quad A_p^+ := \sum_j a_j \frac{(j-1)^p}{p!}, \quad C_p^+ := \sum_j c_j \frac{(j-1)^p}{p!}, \quad \gamma_p^+ := A_p^+ - C_{p-1}^+,$$

the same computation as (3.8)–(3.10) gives

$$(3.13) \quad \tau^w = \sum_{p > p_*} \Delta T^p \gamma_p^+ \tilde{N}_{n+1}^{(p-1)}, \quad \tilde{N}_{n+1}^{(m)} := \left. \frac{d^m \tilde{N}}{dt^m} \right|_{t_{n+1}} = E^{-1} \sum_{j=0}^m \binom{m}{j} (-L)^j N^{(m-j)}(t_{n+1}),$$

The last identity follows by applying the Leibniz rule to (2.6) at t_{n+1} , where the exponential factor is E^{-1} . Both representations vanish through the same order; in

particular, $\gamma_p^+ = 0$ for $p \leq p_*$ if and only if $\gamma_p = 0$ for $p \leq p_*$, because both series expand the same function of ΔT . Their higher-order constants nevertheless differ, and for the family (3.4) they are

$$(3.14) \quad \gamma_3 = \gamma_3^+ = \frac{3\gamma + 1}{3(2\gamma + 1)}, \quad \gamma_4 = \frac{-\gamma}{6(2\gamma + 1)}, \quad \gamma_4^+ = \frac{-(7\gamma + 2)}{6(2\gamma + 1)}.$$

Table 3.2 reports representative values. The pair (γ_p, γ_p^+) is the exponential-integrator analogue of the Adams–Bashforth and Adams–Moulton error constants.

TABLE 3.2
Current-level and new-level error constants.

Scheme	Orders p	γ_p	γ_p^+
IF-Euler	2, 3, 4	$(\frac{1}{2}, \frac{1}{6}, \frac{1}{24})$	$(\frac{1}{2}, -\frac{1}{3}, \frac{1}{8})$
IF-AB2	3, 4	$(\frac{5}{12}, -\frac{1}{24})$	$(\frac{5}{12}, -\frac{11}{24})$

3.3. Stability analysis. Let $\bar{u}(t)$ solve (2.1) and $v := u - \bar{u}$. To first order in v ,

$$(3.15) \quad \dot{v} = L v + J(t) v, \quad J(t) := N'(\bar{u}(t)).$$

No restriction is imposed on N'' ; in particular, $J(t)$ varies along the reference trajectory $\bar{u}$.

Set $\eta(t) := J(t)v(t)$, the linearization of $N(u(t))$ about $\bar{u}$, and write $\eta^{(k)} := \frac{d^k}{dt^k} \eta|_{t_n}$. Differentiating (3.15) and applying the Leibniz rule to $J(t)v(t)$ gives

$$(3.16) \quad v^{(k+1)} = L v^{(k)} + \eta^{(k)}, \quad \eta^{(k)} = \sum_{j=0}^k \binom{k}{j} J^{(j)} v^{(k-j)}, \quad J^{(j)} = \left. \frac{d^j}{dt^j} N'(\bar{u}(t)) \right|_{t_n}.$$

The term $J^{(k)}v$ appears because η is a product, whereas $N(u(t))$ is a composition; for example, $\eta^{(1)} = J^{(0)}v^{(1)} + J^{(1)}v$ is the linearization of $N^{(1)} = N'(u)u^{(1)}$ and equals $N'(\bar{u})v^{(1)} + N''(\bar{u})[v, \bar{u}^{(1)}]$. More generally, $J^{(j)} \neq 0$ for $j \geq 1$ whenever $\bar{u}$ evolves, so $\eta^{(k)}$ is the linearization of $N^{(k)}$ about $\bar{u}$.

The transformed perturbation is defined as in (2.6)–(2.9):

$$(3.17) \quad \omega(t) := e^{-(t-t_n)L} v(t), \quad \dot{\omega} = \tilde{\eta}(t) := e^{-(t-t_n)L} \eta(t), \quad \tilde{\eta}^{(m)} = \sum_{j=0}^m \binom{m}{j} (-L)^j \eta^{(m-j)}.$$

Hold $J^{(0)}, \dots, J^{(S-1)}$ at their values at t_n and treat (3.16) as a constant-coefficient system (the frozen-coefficient analysis of [56]). For one spatial mode, with the base rungs $\bar{u}^{(m)}$ uniform in space, L and each $J^{(j)}$ become multipliers:

$$(3.18) \quad L \rightarrow \lambda_L, \quad J^{(j)} \rightarrow \lambda_{N,j}, \quad z_L := \Delta T \lambda_L, \quad z_{N,j} := \Delta T^{j+1} \lambda_{N,j} \quad (j = 0, \dots, S-1).$$

Here $\lambda_{N,0}$ is the Jacobian of N at the current state and $\lambda_{N,j \geq 1}$ are its time derivatives along $\bar{u}$. Write

$$(3.19) \quad v^{(k)} \rightarrow \Delta T^{-k} \hat{v}_k v, \quad \eta^{(k)} \rightarrow \Delta T^{-(k+1)} \hat{\eta}_k v, \quad \tilde{\eta}^{(k)} \rightarrow \Delta T^{-(k+1)} \tilde{\eta}_k v.$$

Multiplying (3.16) by the appropriate powers of ΔT ,

$$(3.20) \quad \boxed{\hat{v}_0 = 1, \quad \hat{v}_{k+1} = z_L \hat{v}_k + \hat{\eta}_k, \quad \hat{\eta}_k = \sum_{j=0}^k \binom{k}{j} z_{N,j} \hat{v}_{k-j},}$$

so that $\hat{\eta}_0 = z_{N,0}$ and all of these quantities are polynomials in $(z_L, z_{N,0}, \dots, z_{N,S-1})$. From (3.17),

$$(3.21) \quad \tilde{\eta}_m = \sum_{j=0}^m \binom{m}{j} (-z_L)^j \tilde{\eta}_{m-j}.$$

LEMMA 3.1. $\tilde{\eta}_m = \hat{\eta}_m|_{z_L=0}$; in particular $\tilde{\eta}_m$ does not depend on z_L .

Proof. For an individual mode, (3.17) becomes $\dot{\omega} = \tilde{\eta}$ and $\tilde{\eta}(t) = \lambda_J(t)\omega(t)$, where $\lambda_J(t) := \sum_j \lambda_{N,j}(t - t_n)^j/j!$ is the frozen multiplier of $J(t)$ and $\omega(t_n) = v$. This is the modewise form of (3.15) with $\lambda_L = 0$; hence, the derivatives of ω and $\tilde{\eta}$ at t_n satisfy (3.20) with $z_L = 0$. Equation (3.21) provides the same derivatives by direct transformation, which proves the assertion. $\square$

At the operator level, the commutators $[J^{(j)}, L]$ carry the L -dependence of the w -dynamics, but they vanish when L and the $J^{(j)}$ share an eigenbasis, as they do for each mode. The modewise reduction therefore reflects the problem structure rather than the numerical scheme.

Because the scheme (3.3) evaluates only $N(u^{n+j})$, its linearization is $N(u^{n+j}) - N(\bar{u}(t_{n+j})) \rightarrow J(t_{n+j})v^{n+j}$. Freezing this coefficient at t_n gives $J(t_{n+j}) \rightarrow \lambda_{N,0}$, so that only $z_{N,0}$ enters; for each mode, moreover, $E \rightarrow e^{z_L}$. Substituting $v^{n+j} = g^j v^n$ into (3.3), we write

$$(3.22) \quad g =: e^{z_L} g_w,$$

Then $e^{-jz_L} g^j = g_w^j$ represents the transported perturbation history, and

$$(3.23) \quad g v^n = e^{z_L} \left[\sum_{j \leq 0} (-a_j + c_j z_{N,0}) g_w^j \right] v^n \iff g_w = \sum_{j \leq 0} (-a_j + c_j z_{N,0}) g_w^j.$$

After clearing negative powers,

$$(3.24) \quad \pi_w(g_w; z_{N,0}) = \sum_{j=-(s-1)}^1 a_j g_w^{j+s-1} - z_{N,0} \sum_{j=-(s-1)}^0 c_j g_w^{j+s-1} = 0, \quad g = e^{z_L} g_w.$$

This degree- s polynomial is independent of z_L and coincides with the companion polynomial of the explicit method (a, c) applied to $\dot{\omega} = \lambda_{N,0}\omega$. Absolute stability therefore requires every root $g = e^{z_L} g_w$ to lie in the closed unit disk, with roots on the unit circle remaining simple [2]:

$$(3.25) \quad |g| \leq 1 \iff |g_w| \leq e^{-\operatorname{Re} z_L}.$$

The flow of the frozen problem multiplies each mode by $e^{z_L + z_{N,0}}$ per step; the flow-relative criterion is

$$(3.26) \quad |g| \leq |e^{z_L + z_{N,0}}| \iff |g_w| \leq |e^{z_{N,0}}|,$$

in which z_L cancels. The principal root satisfies

$$(3.27) \quad g_{w,1} = e^{z_{N,0}} + \mathcal{O}(z_{N,0}^{p_*+1}), \quad g_1 = e^{z_L} g_{w,1} = e^{z_L + z_{N,0}} + \mathcal{O}(z_{N,0}^{p_*+1}).$$

The higher multipliers $z_{N,j \geq 1}$ do not enter the base scheme; they enter through the correction (Section 3.4) and the reconstructed ladders (Section 4 onward).

Linear N . In this case $N'' \equiv 0$, so $J^{(j \geq 1)} \equiv 0$ exactly and $z_{N,j \geq 1} = 0$. Then (3.20) gives, with $z := z_L + z_{N,0}$,

$$(3.28) \quad \hat{v}_k = z^k, \quad \hat{\eta}_k = z_{N,0} z^k, \quad \tilde{\eta}_k = z_{N,0}^{k+1},$$

which is the classical split test problem; $\tilde{\eta}_k$ is its z_L -free image.

Quadratic N (Section 2.3). By (2.12), $J^{(j)} = 2P(\bar{u}^{(j)}, \cdot)$; with $\bar{u}^{(j)}$ uniform and μ_P the per-mode symbol of P with one slot frozen, $z_{N,j} = 2\Delta T^{j+1} \bar{u}^{(j)} \mu_P$. Thus all S multipliers are one kernel symbol times the base flow's own rungs, and (3.20) is a polynomial in $(z_L, \Delta T \mu_P)$.

3.4. Higher-order correction. For a target order $q > p_*$, define the correction in the integrating-factor variable as

$$(3.29) \quad \delta := \sum_{p=p_*+1}^q \Delta T^p \tau_p^w = \sum_{p=p_*+1}^q \Delta T^p \gamma_p \tilde{N}^{(p-1)}.$$

We add δ to w^{n+1} before transforming back to the original variable. This gives the explicit correction

$$(3.30) \quad u^{n+1} = E \left[\sum_{j \leq 0} (-a_j) E^{-j} u^{n+j} + \Delta T \sum_{j \leq 0} c_j E^{-j} N(u^{n+j}) + \sum_{p=p_*+1}^q \Delta T^p \gamma_p \tilde{N}^{(p-1)} \right].$$

By (3.10) and (3.2), the augmented local error is

$$(3.31) \quad \tau_{\text{aug}} := u(t_{n+1}) - E[w^{n+1} + \delta] = E[\tau^w - \delta] = \Delta T^{q+1} E \tau_{q+1}^w + \mathcal{O}(\Delta T^{q+2}).$$

Thus, the correction removes all LMM local-error terms through order q , leaving the order- $(q+1)$ coefficient $E \tau_{q+1}^w$ of the underlying exponential multistep formula as the leading residual. In this way, the construction defines both the correction and its target order without an external reference integrator.

Implicitly weighted correction. Equation (3.13) expresses the same local error as $\sum_{p>p_*} \Delta T^p \gamma_p^+ \tilde{N}_{n+1}^{(p-1)}$. Any convex combination of the two representations remains an exact representation. For $\theta \in [0, 1]$, define

$$(3.32) \quad \delta_\theta := \theta \sum_{p=p_*+1}^q \Delta T^p \gamma_p \tilde{N}^{(p-1)} + (1-\theta) \sum_{p=p_*+1}^q \Delta T^p \gamma_p^+ \tilde{N}_{n+1}^{(p-1)}, \quad \delta_1 = \delta.$$

Inserting (3.13) and multiplying by E , the factor E^{-1} of the new-level rungs cancels, so that the corrected step reads

$$(3.33) \quad u^{n+1} = E \left[w^{n+1} + \theta \sum_{p=p_*+1}^q \Delta T^p \gamma_p \tilde{N}^{(p-1)} \right] + (1-\theta) \sum_{p=p_*+1}^q \Delta T^p \gamma_p^+ \tilde{N}^{(p-1)}(u^{n+1}),$$

Here $\tilde{N}^{(m)}(u^{n+1}) := \sum_{j=0}^m \binom{m}{j} (-L)^j N^{(m-j)}(t_{n+1})$ denotes the untransported rung ladder at the new level. Because these rungs depend on the unknown u^{n+1} , equation (3.33) is implicit and has the augmented local error

$$(3.34) \quad \tau_{\text{aug}, \theta} = \theta \Delta T^{q+1} E \gamma_{q+1} \tilde{N}^{(q)} + (1-\theta) \Delta T^{q+1} \gamma_{q+1}^+ \tilde{N}^{(q)}(u(t_{n+1})) + \mathcal{O}(\Delta T^{q+2}),$$

This error is of order q for every θ , because both representations are truncated at the same order.

Linearly implicit realization. Let $\tilde{u}^{n+1} := E[w^{n+1} + \delta]$ denote the value of the explicit correction (3.30), used as a predictor in the sense of the predictor–corrector methods [57, 2], with local error $\mathcal{O}(\Delta T^{q+1})$. For each rung fix a linear operator $\mathcal{J}_m$ (frozen within the step) and split

$$(3.35) \quad \tilde{N}^{(m)}(u^{n+1}) = \mathcal{J}_m u^{n+1} + [\tilde{N}^{(m)}(\tilde{u}^{n+1}) - \mathcal{J}_m \tilde{u}^{n+1}] + \mathcal{O}(\|u^{n+1} - \tilde{u}^{n+1}\|).$$

Substituting (3.35) into (3.33) and dropping the last term yields the linearly implicit step

$$(3.36) \quad \left[I - (1-\theta) \sum_{p=p_*+1}^q \Delta T^p \gamma_p^+ \mathcal{J}_{p-1} \right] u^{n+1} = E \left[w^{n+1} + \theta \sum_p \Delta T^p \gamma_p \tilde{N}^{(p-1)} \right] + (1-\theta) \sum_p \Delta T^p \gamma_p^+ [\tilde{N}^{(p-1)}(\tilde{u}^{n+1}) - \mathcal{J}_{p-1} \tilde{u}^{n+1}].$$

For $p \geq p_\star + 1 \geq 2$, the discarded term is $\mathcal{O}(\Delta T^p \Delta T^{q+1}) = \mathcal{O}(\Delta T^{q+3})$, two orders smaller than (3.34). Consequently, (3.36) has order q for every bounded choice of $\mathcal{J}_m$, although this choice affects the stability region analyzed in Section 3.5.

When $\mathcal{J}_m$ shares an eigenbasis with L , the left-hand operator is diagonal in that basis, and the solve requires one division per mode at the cost of one application of E . Relative to (3.30), the method evaluates one additional rung ladder at $\tilde{u}^{n+1}$ and performs one diagonal solve, while forming neither a nonlinear system nor the Jacobian of N . For quadratic N and $q = 4$, this additional ladder requires nine bilinear products.

The reconstruction in Section 4 replaces the ladders with finite differences and one surrogate action, leaving one evaluation of N at $\tilde{u}^{n+1}$ as the remaining overhead. The construction follows the linearly implicit treatment of stiff terms [13, 9] and the alternating implicit–explicit splitting of [10], so that only the component that is inexpensive to invert remains implicit.

By (2.9), the correction requires $N(u)$, which is already evaluated by the base scheme, the nonlinear time derivatives $N^{(1)}, \dots, N^{(q-1)}$, and their products with powers of L ; only the time derivatives are unavailable during time integration, and no term $L^k u$ occurs.

3.5. Stability of the corrected method. Linearize (3.30) about $\bar{u}$. The correction is a function of $N^{(0)}, \dots, N^{(q-1)}$; its linearization replaces $N^{(m)}$ by $\eta^{(m)}$ (Section 3.3), and hence $\tilde{N}^{(p-1)}$ by $\tilde{\eta}^{(p-1)}$ of (3.17). Per mode, by (3.19),

$$(3.37) \quad \Delta T^p \gamma_p \tilde{\eta}^{(p-1)} \rightarrow \gamma_p \tilde{\eta}_{p-1} v, \quad \hat{\kappa} := \sum_{p=p_\star+1}^q \gamma_p \tilde{\eta}_{p-1}.$$

By Lemma 3.1, $\hat{\kappa}$ depends on $(z_{N,0}, \dots, z_{N,q-2})$ only. With $v^{n+j} = g^j v^n$ and (3.22), the linearized corrected step reads

$$(3.38) \quad g_w = \sum_{j \leq 0} (-a_j + c_j z_{N,0}) g_w^j + \hat{\kappa},$$

so the corrected companion polynomial is

$$(3.39) \quad \boxed{\pi_w^\delta(g_w) := \pi_w(g_w; z_{N,0}) - \hat{\kappa} g_w^{s-1} = 0, \quad g^\delta := e^{z_L} g_w^\delta,}$$

with g_w^δ a root of π_w^δ . Because π_w^δ is z_L -free, z_L enters g^δ only through the exact factor e^{z_L} ; moreover, since the corrected scheme has local error $\mathcal{O}(\Delta T^{q+1})$ in w , its principal root is the Taylor polynomial of ω :

$$(3.40) \quad g_{w,1}^\delta = \sum_{k=0}^q \frac{\hat{\nu}_k|_{z_L=0}}{k!} + \mathcal{O}(\Delta T^{q+1}).$$

Because π_w^δ has degree s in g_w and polynomial coefficients in $z_{N,0}$, the stability region of the explicit correction is bounded. For $s = 1$, it is the region associated with the degree- q Taylor polynomial of $e^{z_{N,0}}$.

Implicitly weighted correction. Linearize (3.33). The new-level rungs $\check{N}^{(p-1)}(u^{n+1})$ form the same ladder one step later. With frozen coefficients, their modewise linearization is the symbol $\tilde{\eta}_{p-1}$ from (3.21), applied to $v^{n+1} = g v^n$. Lemma 3.1 remains valid because the transport factors cancel as in (3.33). Define

$$(3.41) \quad \hat{\kappa}^+ := \sum_{p=p_\star+1}^q \gamma_p^+ \tilde{\eta}_{p-1},$$

the linearized step is $g_w = \sum_{j \leq 0} (-a_j + c_j z_{N,0}) g_w^j + \theta \hat{\kappa} + (1 - \theta) \hat{\kappa}^+ g_w$, and

$$(3.42) \quad \boxed{\pi_w^\theta(g_w) := \pi_w(g_w; z_{N,0}) - \theta \hat{\kappa} g_w^{s-1} - (1 - \theta) \hat{\kappa}^+ g_w^s = 0, \quad g^\theta := e^{z_L} g_w^\theta.}$$

Because the implicit term changes the leading coefficient to $1 - (1 - \theta) \hat{\kappa}^+$, the roots are rational functions of the multipliers; for $s = 1$,

$$(3.43) \quad g_w^\theta = \frac{1 + z_{N,0} + \theta \hat{\kappa}}{1 - (1 - \theta) \hat{\kappa}^+},$$

and for linear N , by (3.28), $\hat{\kappa} = \sum_p \gamma_p z_{N,0}^p$ and $\hat{\kappa}^+ = \sum_p \gamma_p^+ z_{N,0}^p$; for $q = 2$ and $\theta = \frac{1}{2}$ this is $(1 + z_{N,0}/2)/(1 - z_{N,0}/2)$, the amplification factor of the A-stable trapezoidal rule [11, 12].

PROPOSITION 3.2. *Let $s = 1$, N linear, and $\gamma_q^+ \neq 0$. Then $|g_w^\theta| \rightarrow \theta |\gamma_q| / ((1 - \theta) |\gamma_q^+|)$ as $|z_{N,0}| \rightarrow \infty$; the stability region $\{|g_w^\theta| \leq 1\}$ contains a neighborhood of infinity if and only if*

$$(3.44) \quad \theta |\gamma_q| < (1 - \theta) |\gamma_q^+|.$$

Proof. Numerator and denominator of (3.43) are polynomials of degree q in $z_{N,0}$ with leading coefficients $\theta \gamma_q$ and $-(1 - \theta) \gamma_q^+$. $\square$

For IF-Euler with $q = 4$, condition (3.44) becomes $\theta < \frac{3}{4}$. The region is then unbounded in the left half-plane and contains the imaginary axis. Isolated poles occur where the denominator of (3.43) vanishes. For $s = 2$, the leading coefficient of (3.42) has the same form and scales the parasitic root by the same factor. The family (3.4) with $q = 4$ and $\theta = \frac{1}{4}$ has an unbounded region for $\gamma \geq \frac{1}{2}$. Bounded instability pockets remain near the negative real axis; see Section 5.

The linearly implicit realization (3.36) reproduces (3.42) when the modewise symbol of $\mathcal{J}_{p-1}$ equals $\Delta T^{-p} \tilde{\eta}_{p-1}$. A symbol mismatch introduces the predictor and produces a bounded region, although the companion polynomial retains degree s and (3.35) preserves the order. The supplementary materials quantify the numerical sensitivity to a representative symbol mismatch.

For quadratic N , $\hat{\kappa}$ is a polynomial of degree q in the kernel symbol $\Delta T \mu_P$ with coefficients depending on the base rungs only.

4. S -stage d -multilinear convolution approximation of higher-order derivatives. In the remainder of the analysis, every error is defined as the approximate quantity minus its exact counterpart.

At step n , the stored values, uniformly spaced by ΔT , are

$$(4.1) \quad u(t_{n-i}), \quad N(u(t_{n-i})), \quad i = 0, 1, \dots, S-1.$$

The surrogate receives exclusively these stored histories: neither transported fields $E^i u(t_{n-i})$ nor products involving L or E are included in its input, and the time derivatives $u^{(m)}$ and $N^{(m)}$ for $m \geq 1$, including $u^{(S)}$ and $N^{(S)}$, are not assumed to be available.

4.1. Finite-difference reconstruction. For each $m \in \{0, \dots, S-1\}$, the weights W_{mi} , $i = 0, \dots, S-1$, are defined by

$$(4.2) \quad \sum_{i=0}^{S-1} W_{mi} \frac{(-i)^p}{p!} = \delta_{mp}, \quad p = 0, 1, \dots, S-1.$$

The nodes $0, -1, \dots, -(S-1)$ are distinct, so (4.2) has a unique solution. Define

$$(4.3) \quad u_{\text{FD}}^{(m)} := \frac{1}{\Delta T^m} \sum_{i=0}^{S-1} W_{mi} u(t_{n-i}), \quad N_{\text{FD}}^{(m)} := \frac{1}{\Delta T^m} \sum_{i=0}^{S-1} W_{mi} N(u(t_{n-i})).$$

Assume $u \in C^{S+2}$, expand $u(t_{n-i})$ about t_n , multiply by W_{mi} , and sum over i . Equation (4.2) then reduces the terms with $p \leq S-1$ to δ_{mp} , while the next two terms define the finite-difference moments:

$$(4.4) \quad \sum_i W_{mi} u(t_{n-i}) = \Delta T^m u^{(m)} + K_m \Delta T^S u^{(S)} + K_m^{(2)} \Delta T^{S+1} u^{(S+1)} + \mathcal{O}(\Delta T^{S+2}),$$

$$K_m := \frac{1}{S!} \sum_{i=0}^{S-1} W_{mi} (-i)^S, \quad K_m^{(2)} := \frac{1}{(S+1)!} \sum_{i=0}^{S-1} W_{mi} (-i)^{S+1}.$$

Table 4.1 gives the moments for $S = 4$. Dividing by ΔT^m gives

TABLE 4.1
Finite-difference moments for a four-level history.

m	K_m	$K_m^{(2)}$
0	0	0
1	$-\frac{1}{4}$	$\frac{3}{10}$
2	$-\frac{11}{12}$	$\frac{1}{4}$
3	$-\frac{3}{2}$	$\frac{5}{4}$

$$(4.5) \quad \boxed{u_{\text{FD}}^{(m)} - u^{(m)} = K_m \Delta T^{S-m} u^{(S)} + K_m^{(2)} \Delta T^{S-m+1} u^{(S+1)} + \mathcal{O}(\Delta T^{S-m+2}), \quad m < S.}$$

Since $t \mapsto N(u(t))$ is C^{S+1} , (4.5) applies with $u \rightarrow N(u)$ and $u^{(p)} \rightarrow N^{(p)}$:

$$(4.6) \quad \boxed{N_{\text{FD}}^{(m)} - N^{(m)} = K_m \Delta T^{S-m} N^{(S)} + K_m^{(2)} \Delta T^{S-m+1} N^{(S+1)} + \mathcal{O}(\Delta T^{S-m+2}), \quad m < S.}$$

Since $W_{0i} = \delta_{i0}$ by uniqueness, $K_0 = 0$, $u_{\text{FD}}^{(0)} = u$, and $N_{\text{FD}}^{(0)} = N(u)$.

4.2. Surrogate approximation and recursive construction. Equations (4.5)–(4.6) express every finite-difference residual as a known factor $K_m \Delta T^{S-m}$ times either $u^{(S)}$ or $N^{(S)}$. Since the identity $u^{(S)} = Lu^{(S-1)} + N^{(S-1)}$ resolves the state derivative from level $S-1$, the single field $N^{(S)}$ closes the hierarchy. The target is a derivative of $N(u(t))$, rather than of $\tilde{N}(t)$, and (2.9) is therefore applied only after the slots have been corrected.

Setting $k = S-1$ in (2.5) gives

$$(4.7) \quad \boxed{N^{(S)} = \sum_{j=0}^{S-1} \binom{S-1}{j} J^{(j)} u^{(S-j)}.$$

The top rung inside the $j = 0$ term is supplied by (2.3) at $k = S-1$:

$$(4.8) \quad u^{(S)} = Lu^{(S-1)} + N^{(S-1)}.$$

The $j = 0$ term is the Jacobian acting on the top rung; the $j \geq 1$ terms are b_S of (2.2).

The N -history is read by finite differences up to level $S-1$, and one surrogate supplies level S (deeper recursions, in which the surrogate supplies several levels, follow the same pattern and are not needed here). The reconstructed rungs are

$$(4.9) \quad \hat{u}^{(m)} := u_{\text{FD}}^{(m)}, \quad 1 \leq m \leq S-1,$$

$$(4.10) \quad \hat{u}^{(S)} := Lu_{\text{FD}}^{(S-1)} + N_{\text{FD}}^{(S-1)}.$$

For $j = 0, \dots, S-1$, let $\hat{J}^{(j)} : \mathbb{R}^M \rightarrow \mathbb{R}^M$ be a linear map held fixed within a time step, with no additional structural assumption imposed at this stage. Substituting (4.9)–(4.10) and these surrogates into (4.7) gives

$$(4.11) \quad \hat{N}^{(S)} := \sum_{j=0}^{S-1} \binom{S-1}{j} \hat{J}^{(j)} \hat{u}^{(S-j)}.$$

4.3. Error analysis. Define the errors

$$(4.12) \quad \varepsilon_m^u := \hat{u}^{(m)} - u^{(m)}, \quad \varepsilon_m^N := \hat{N}^{(m)} - N^{(m)}, \quad D^{(j)} := \hat{J}^{(j)} - J^{(j)}.$$

From (4.5)–(4.6), for the stencil-supplied rungs,

$$(4.13) \quad \varepsilon_m^u = K_m \Delta T^{S-m} u^{(S)} + \mathcal{O}(\Delta T^{S-m+1}), \quad 1 \leq m \leq S-1, \quad \varepsilon_{S-1}^N = K_{S-1} \Delta T N^{(S)} + \mathcal{O}(\Delta T^2).$$

For linear maps $\hat{J}$, J and vectors $\hat{u}$, u , use $\hat{J}\hat{u} - Ju = \hat{J}(\hat{u} - u) + (\hat{J} - J)u$. With (4.12), this identity becomes

$$(4.14) \quad \hat{J}^{(j)} \hat{u}^{(m)} - J^{(j)} u^{(m)} = D^{(j)} u^{(m)} + \hat{J}^{(j)} \varepsilon_m^u.$$

This identity is exact and uses only the linearity of $\hat{J}^{(j)}$. Subtracting (2.3) from (4.10) and (2.5) from (4.11), and applying (4.14) to each summand, gives

$$(4.15) \quad \varepsilon_S^u = L \varepsilon_{S-1}^u + \varepsilon_{S-1}^N, \quad \varepsilon_S^N = \sum_{j=0}^{S-1} \binom{S-1}{j} \left[D^{(j)} u^{(S-j)} + \hat{J}^{(j)} \varepsilon_{S-j}^u \right],$$

linear in the errors and in the $D^{(j)}$. With (4.13) and (2.3) at $k = S$,

$$(4.16) \quad \varepsilon_S^u = K_{S-1} \Delta T (Lu^{(S)} + N^{(S)}) + \mathcal{O}(\Delta T^2) = K_{S-1} \Delta T u^{(S+1)} + \mathcal{O}(\Delta T^2),$$

which is where $N \in C^{S+1}$ is used. In the second sum of (4.15), the $j = 0$ term carries (4.16), the $j = 1$ term carries ε_{S-1}^u from (4.13), and the terms $j \geq 2$ are $\mathcal{O}(\Delta T^2)$ because $\varepsilon_{S-j}^u = \mathcal{O}(\Delta T^j)$. Hence

$$(4.17) \quad \varepsilon_S^N = \hat{N}^{(S)} - N^{(S)} = \underbrace{\sum_{j=0}^{S-1} \binom{S-1}{j} D^{(j)} u^{(S-j)}}_{\mathcal{O}(\Delta T^0)} + \underbrace{\Delta T K_{S-1} \left[\hat{J}^{(0)} u^{(S+1)} + (S-1) \hat{J}^{(1)} u^{(S)} \right]}_{\mathcal{O}(\Delta T^1)} + \mathcal{O}(\Delta T^2).$$

For $m = 1, \dots, q-1$, rearranging (4.6) with $N^{(S)}$ replaced by its prediction defines the corrected slots

$$(4.18) \quad \hat{N}^{(m)} := N_{\text{FD}}^{(m)} - K_m \Delta T^{S-m} \hat{N}^{(S)}.$$

Subtracting $N^{(m)}$ and inserting (4.6), the $N^{(S)}$ terms cancel exactly:

$$(4.19) \quad \hat{N}^{(m)} - N^{(m)} = -K_m \Delta T^{S-m} \varepsilon_S^N + K_m^{(2)} \Delta T^{S-m+1} N^{(S+1)} + \mathcal{O}(\Delta T^{S-m+2}), \quad 1 \leq m \leq q-1.$$

The second term is the part of the finite-difference remainder that the prediction of $N^{(S)}$ does not remove; it is of the same order as the leak contribution to ε_S^N in (4.17) and is carried below. The $m = 0$ slot is exact, $\hat{N}^{(0)} := N(u)$, consistent with $K_0 = 0$ in (4.1).

Applying the binomial map (2.9) to the corrected slots, with $\hat{N}^{(0)} := N(u)$, gives

$$(4.20) \quad \hat{N}^{(m)} := \sum_{j=0}^m \binom{m}{j} (-L)^j \hat{N}^{(m-j)}, \quad m = 0, \dots, q-1.$$

This is the first point downstream of the surrogate where L acts on a surrogate-dependent quantity. From (3.29),

$$(4.21) \quad \delta = \sum_{p=p_*+1}^q \Delta T^p \gamma_p \hat{N}^{(p-1)} = \sum_{p=p_*+1}^q \Delta T^p \gamma_p \sum_{j=0}^{p-1} \binom{p-1}{j} (-L)^j N^{(p-1-j)}.$$

Replacing $\hat{N}^{(p-1)}$ by (4.20) defines the learned correction

$$(4.22) \quad \hat{\delta} := \sum_{p=p_*+1}^q \Delta T^p \gamma_p \hat{\hat{N}}^{(p-1)}, \quad \hat{u}^{n+1} := E[w^{n+1} + \hat{\delta}],$$

with w^{n+1} the base value (3.2). Subtracting (4.21) from (4.22), the $j = p-1$ term is the exact $m = 0$ slot and drops out:

$$(4.23) \quad \hat{\delta} - \delta = \sum_{p=p_*+1}^q \Delta T^p \gamma_p \sum_{j=0}^{p-2} \binom{p-1}{j} (-L)^j (\hat{N}^{(p-1-j)} - N^{(p-1-j)}).$$

Inserting (4.19) with $m = p-1-j \in \{1, \dots, p-1\}$ gives

$$(4.24) \quad \begin{aligned} \hat{\delta} - \delta &= \sum_{p=p_*+1}^q \Delta T^p \gamma_p \sum_{j=0}^{p-2} \binom{p-1}{j} (-L)^j \left[-K_{p-1-j} \Delta T^{S-p+1+j} \varepsilon_S^N \right. \\ &\quad \left. + K_{p-1-j}^{(2)} \Delta T^{S-p+2+j} N^{(S+1)} + \mathcal{O}(\Delta T^{S-p+3+j}) \right] \\ &= -\Delta T^{S+1} \sum_{p=p_*+1}^q \gamma_p \sum_{j=0}^{p-2} \binom{p-1}{j} K_{p-1-j} (-\Delta T L)^j \varepsilon_S^N \\ (4.25) \quad &\quad + \Delta T^{S+2} \sum_{p=p_*+1}^q \gamma_p \sum_{j=0}^{p-2} \binom{p-1}{j} K_{p-1-j}^{(2)} (-\Delta T L)^j N^{(S+1)} + \mathcal{O}(\Delta T^{S+3}). \end{aligned}$$

In (4.24), the exponent of ΔT is $p + S - p + 1 + j = S + 1 + j$; hence, every slot enters at order $S+1$ or higher, with L^j carrying the excess j , and the remainder begins at order $S+3$. We therefore define the polynomials

$$(4.26) \quad \tilde{K}_m(x) := \sum_{j=0}^{m-1} \binom{m}{j} K_{m-j} x^j, \quad \tilde{K}_m^{(2)}(x) := \sum_{j=0}^{m-1} \binom{m}{j} K_{m-j}^{(2)} x^j, \quad \tilde{K}_m(0) = K_m,$$

of degree $m-1$ ($K_0 = K_0^{(2)} = 0$ removes $j = m$), and

$$(4.27) \quad \Theta(x) := \sum_{p=p_*+1}^q \gamma_p \tilde{K}_{p-1}(x), \quad \Theta^{(2)}(x) := \sum_{p=p_*+1}^q \gamma_p \tilde{K}_{p-1}^{(2)}(x), \quad \Theta(0) = \sum_{p=p_*+1}^q \gamma_p K_{p-1},$$

polynomials of degree $q-2$. Then (4.25) reads

$$(4.28) \quad \boxed{\hat{\delta} = \delta - \Delta T^{S+1} \Theta(-\Delta T L) \varepsilon_S^N + \Delta T^{S+2} \Theta^{(2)}(-\Delta T L) N^{(S+1)} + \mathcal{O}(\Delta T^{S+3}).}$$

With (3.31), $\hat{\tau} := u(t_{n+1}) - \hat{u}^{n+1} = E[\tau^w - \delta] - E(\hat{\delta} - \delta)$; the $\mathcal{O}(\Delta T^0)$ part of ε_S^N in (4.17) lands at ΔT^{S+1} , the $\mathcal{O}(\Delta T^1)$ part at ΔT^{S+2} , and the $\mathcal{O}(\Delta T^2)$ part at ΔT^{S+3} .

(4.29)

$$\begin{aligned}
\hat{\tau} = & \Delta T^{q+1} E \gamma_{q+1} \tilde{N}^{(q)} \\
& + \Delta T^{S+1} E \Theta(-\Delta T L) \sum_{j=0}^{S-1} \binom{S-1}{j} D^{(j)} u^{(S-j)} \\
& + \Delta T^{S+2} E \left\{ \Theta(-\Delta T L) K_{S-1} \left[\hat{J}^{(0)} u^{(S+1)} + (S-1) \hat{J}^{(1)} u^{(S)} \right] - \Theta^{(2)}(-\Delta T L) N^{(S+1)} \right\} \\
& + \mathcal{O}(\Delta T^{q+2}, \Delta T^{S+3}).
\end{aligned}$$

The surrogate term does not lower the prescribed target order if and only if $S+1 \geq q+1$, that is,

(4.30)

$$S \geq q.$$

Because level m enters δ only through $\tilde{N}^{(p-1)}$ with $p \geq m+1$, it carries at least the factor ΔT^{m+1} and therefore requires accuracy $\mathcal{O}(\Delta T^{q-m})$. The finite differences provide $\mathcal{O}(\Delta T^{S-m})$, so (4.30) follows for every m and the finite-difference ladder retains order q whenever $S \geq q$. Within this ladder, the surrogate replaces $\Theta(-\Delta T L) N^{(S)}$ by $\Theta(-\Delta T L) \varepsilon_S^N$.

The preceding results use the linearity of $\hat{J}^{(j)}$ only in (4.14) and its fixedness within a step only in (4.15). Likewise, the regularity condition $N \in C^{S+1}$ enters through (4.4) and (4.16), while the history-depth condition $S \geq q$ is invoked in (4.30). Applications of E^j are required only in (3.3) and (4.22), and no particular parameterization of $\hat{J}^{(j)}$ is required.

Linearize (4.22) about $\bar{u}$ with $\hat{J}^{(j)}$ and $D^{(j)}$ frozen at their values on $\bar{u}$; then ε_S^N is linear in v and, per mode,

$$(4.31) \quad \Delta T^{S+1} \varepsilon_S^N \rightarrow \hat{\varepsilon} v, \quad D^{(j)} \rightarrow \Delta T^{-(j+1)} \hat{d}_j, \quad \hat{J}^{(j)} \rightarrow \Delta T^{-(j+1)} (z_{N,j} + \hat{d}_j).$$

From (4.28) and (3.37), the learned correction has symbol $\hat{\kappa} - \Theta(-z_L) \hat{\varepsilon}$; the surrogate-corrected companion polynomial is therefore

$$(4.32) \quad \hat{\pi}_w(g_w) := \pi_w^\delta(g_w) + \Theta(-z_L) \hat{\varepsilon} g_w^{s-1} = 0, \quad \hat{g} := e^{z_L} \hat{g}_w,$$

with $\hat{g}_w$ a root of $\hat{\pi}_w$. Writing $\hat{g}_w = g_w^\delta + \Delta g_w$ and expanding (4.32) to first order in $\hat{\varepsilon}$, using $\pi_w^\delta(g_w^\delta) = 0$, gives

$$(4.33) \quad \Delta g_w = - \frac{\Theta(-z_L) (g_w^\delta)^{s-1}}{\pi_w^{\delta'}(g_w^\delta)} \hat{\varepsilon}, \quad \hat{g} = e^{z_L} (g_w^\delta + \Delta g_w) + \mathcal{O}(\hat{\varepsilon}^2),$$

Here $\pi_w^{\delta'}$ denotes differentiation with respect to g_w . The multiplier z_L enters Δg_w only through the degree- $(q-2)$ polynomial $\Theta(-z_L)$ and through $\hat{\varepsilon}$, whereas the base and corrected roots are independent of z_L . The triangle inequality then shows that (3.26) holds whenever

$$(4.34) \quad |\hat{\varepsilon}| \leq \epsilon_{\max} := \frac{|\pi_w^{\delta'}(g_w^\delta)| (|e^{z_{N,0}}| - |g_w^\delta|)}{|\Theta(-z_L)| |g_w^\delta|^{s-1}}.$$

Per mode, (4.17) gives $\hat{\varepsilon} = \sum_j \binom{S-1}{j} \hat{d}_j \hat{v}_{S-j} + K_{S-1} [(z_{N,0} + \hat{d}_0) \hat{v}_{S+1} + (S-1)(z_{N,1} + \hat{d}_1) \hat{v}_S] + \mathcal{O}(\Delta T^{S+3})$, so that (4.34) is a linear inequality in $(\hat{d}_0, \dots, \hat{d}_{S-1})$ whose surrogate-independent offset is the finite-difference contribution that remains when the surrogate operators are exact.

4.4. Reconstruction at the new level and parasitic roots.

Learned implicitly weighted correction. The correction (3.33) requires rung ladders at two time levels. We reconstruct the ladder at t_n from (4.1). The same procedure reconstructs the ladder at t_{n+1} from the shifted stencil

$$(4.35) \quad u^{n+1}, u(t_n), \dots, u(t_{n-S+2}), \quad N(u^{n+1}), N(u(t_n)), \dots, N(u(t_{n-S+2})),$$

With t_{n+1} as the newest point, we use the rows W_{mi} and moments $K_m, K_m^{(2)}$ from (4.2)–(4.4), counting i from t_{n+1} , and denote the resulting slots by $\hat{N}_{n+1}^{(m)}$ and their untransported ladder by $\hat{N}_{n+1}^{(m)} := \sum_{j \leq m} \binom{m}{j} (-L)^j \hat{N}_{n+1}^{(m-j)}$. The learned correction is

$$(4.36) \quad \delta_\theta := \theta \sum_{p=p_*+1}^q \Delta T^p \gamma_p \hat{N}^{(p-1)} + (1-\theta) \sum_{p=p_*+1}^q \Delta T^p \gamma_p^+ E^{-1} \hat{N}_{n+1}^{(p-1)},$$

and (3.33)–(3.36) apply with $\check{N}^{(m)}$ replaced by $\hat{N}_{n+1}^{(m)}$. In the linearly implicit realization the newest level of (4.35) is the predictor $\tilde{u}^{n+1}$ everywhere except in the linear part, for which $\mathcal{J}_m$ is taken as the derivative of $\hat{N}_{n+1}^{(m)}$ with respect to its newest input, frozen at the mean state; since every reconstruction operation is linear in that input, this derivative is a fixed linear operator (for translation-equivariant surrogates, a convolution, hence diagonal in Fourier space). Its modewise symbol follows the reconstruction literally: with W_{m0} the newest-point weight of row m ,

$$(4.37) \quad \begin{aligned} \hat{r}_m &:= W_{m0} \quad (m < S), & \hat{r}_S &:= (z_L + z_{N,0}) \hat{r}_{S-1}, \\ \hat{p}_m &:= z_{N,0} \hat{r}_m - K_m z_{N,0} \hat{r}_S, & \hat{j}_m &:= \sum_{j=0}^m \binom{m}{j} (-z_L)^j \hat{p}_{m-j}, \end{aligned}$$

so that the implicit factor of (3.36) is $1 - (1-\theta) \sum_p \gamma_p^+ \hat{j}_{p-1}$ per mode.

Error. Each stencil carries the analysis of Sections 4.3–4.3 with its own histories, and (4.36) is linear in the slots; therefore, with $\varepsilon_{S,n}^N$ and $\varepsilon_{S,n+1}^N$ the top-rung errors of the two stencils and $\Theta^+, \Theta^{(2)+}$ defined as in (4.27) with $\gamma_p \rightarrow \gamma_p^+$,

$$(4.38) \quad \boxed{\begin{aligned} E[\hat{\delta}_\theta - \delta_\theta] &= -\Delta T^{S+1} \left[\theta E \Theta(-\Delta T L) \varepsilon_{S,n}^N + (1-\theta) \Theta^+(-\Delta T L) \varepsilon_{S,n+1}^N \right] \\ &\quad + \Delta T^{S+2} \left[\theta E \Theta^{(2)}(-\Delta T L) N^{(S+1)}(t_n) + (1-\theta) \Theta^{(2)+}(-\Delta T L) N^{(S+1)}(t_{n+1}) \right] \\ &\quad + \mathcal{O}(\Delta T^{S+3}). \end{aligned}}$$

Consequently (4.30) is unchanged: $S \geq q$ preserves order q for every θ . The predictor enters only through (3.35): the difference between the remainder evaluated at $\tilde{u}^{n+1}$ and at $u(t_{n+1})$ is $(1-\theta) \sum_p \Delta T^p \gamma_p^+ \mathcal{O}(\Delta T^{q+1}) = \mathcal{O}(\Delta T^{q+3})$.

Parasitic roots. The surrogate-corrected companion polynomial (4.32) treats the surrogate error as a symbol multiplying the current level. The reconstructed slots, however, are linear combinations of *stored levels*: on the perturbation history $v^{n-i} = g^{-i} v^n$ the rows become tap polynomials $\hat{W}_m(g) = \sum_i W_{mi} g^{-i}$, and the learned scheme is an S -step recurrence whatever the depth s of the base formula. For IF-Euler with $S = q = 4$, linear N , and the explicit correction ($\theta = 1$), collecting the levels gives, to first order in $z_{N,0}$,

$$(4.39) \quad \hat{g}^4 = \left(1 + \frac{55}{24} z_{N,0}\right) \hat{g}^3 - \frac{59}{24} z_{N,0} \hat{g}^2 + \frac{37}{24} z_{N,0} \hat{g} - \frac{9}{24} z_{N,0},$$

the Adams–Bashforth 4 recurrence [2]. Thus, the finite-difference taps turn the one-step base formula into a four-step recurrence whose three parasitic roots satisfy $\hat{g}^3 \approx \frac{9}{24} z_{N,0}$ and therefore $|\hat{g}_{\text{par}}| \approx (\frac{9}{24} |z_{N,0}|)^{1/3}$. Consequently, a parasitic root eventually

leaves the unit disk independently of the correction order, a behavior that is not revealed by the local truncation error because it describes only the principal root [1].

The implicit weight changes these history coefficients: the explicit stencil reads the oldest level t_{n-3} with weight $\theta \frac{9}{24} z_{N,0}$, whereas the shifted stencil (4.35) reads t_{n-2} with the Adams–Moulton weight $(1 - \theta) \frac{1}{24} z_{N,0}$ [2] and incorporates its newest level through the implicit factor. Table 4.2 reports the resulting imaginary-axis limits; the admissible interval on the negative real axis likewise expands as θ decreases.

TABLE 4.2
Largest imaginary-axis value of $|z_{N,0}|$ for which all parasitic roots remain in the unit disk.

Scheme	$\theta = 1$	$\theta = \frac{3}{4}$	$\theta = \frac{1}{2}$	$\theta = \frac{1}{4}$	$\theta = 0$
IF-Euler	0.42	0.54	0.74	1.19	2.54
IF-AB2	0.43	0.51	0.61	0.73	0.87

The remaining imaginary-axis restriction comes from the principal root. For IF-Euler with $S = 4$, the leak term in (4.17) and the second-moment term in (4.38) give $|\hat{g}_1| - 1 \approx +\frac{17}{120} |z_{N,0}|^6$. This anti-dissipative $\mathcal{O}(\Delta T^6)$ error is absorbed by $e^{\text{Re } z_L}$ whenever the linear part provides damping.

Remark 4.1. The Burgers testbed in Section 5 verifies (4.38) on exact levels with the exact multilinear surrogate. The correction gap exhibits the predicted sixth-order decay for $S = 4$. The residual of the two-term formula exhibits seventh-order decay. Omitting $\Theta^{(2)}$ instead leaves an order-one relative residual. The predictor term also exhibits seventh-order decay.

Regularity. All statements through the target order hold for $N \in C^{q+1}$ near the trajectory; the finite-difference estimates and (4.16) require $N \in C^{S+1}$ (and C^{S+2} for the second-moment terms). For the quadratic class the requisite higher state derivatives vanish identically.

4.5. Full numerical scheme. Algorithm 4.1 specifies one step of the linearly implicit, implicitly weighted scheme with one surrogate level; setting $\theta = 1$ recovers the explicit correction and gives $\hat{u}^{n+1} = \tilde{u}^{n+1}$. The reconstruction layer Λ acts on both the stored stencil and the shifted predictor stencil using only the histories of u and $N(u)$, while the exponential operator appears only in the base step, predictor, and final solve. Beyond the base method, the work consists of finite-difference combinations, products with L , $2S$ surrogate actions, one evaluation of N at the predictor, and one frozen linear solve; when $\mathcal{J}$ is diagonal in the eigenbasis of L , this solve reduces to one division per mode.

5. Numerical results. We considered the three testbeds summarized in Table 5.1. The periodic problems were discretized by Fourier collocation [55, 16] with truncation dealiasing [54]; because their mean flow $\bar{u}$ gives $\lambda_{N,0}(k) \approx -ik\bar{u}$, the mode-wise analysis applies directly.

Table 5.2 summarizes the common numerical design. In all experiments, the reconstructed rungs used the exact multilinear surrogate on the finite-difference ladder of the transported stencil described in Section 4.4, while the spectral testbeds used the mean-flow linearization as the frozen operator.

The admissibility criterion is independent of z_L by Lemma 3.1.

Figure 5.1 shows the stability regions for the most dispersive KdV mode. At this value of z_L , the explicit LMM has no stable region and the IMEX method, al-

Algorithm 4.1 One step, implicitly weighted correction

Require: stencil (4.1); rows W_{mi} , constants K_m , (4.2), (4.4); $a_j, c_j, \gamma_p, \gamma_p^+$, (3.8), (3.12); weight θ ; E, E^j , (2.2); surrogates $\hat{J}^{(j)}$; frozen operator $\mathcal{J} := \sum_p \Delta T^p \gamma_p^+ \mathcal{J}_{p-1}$, (3.35)

- 1: **Base step in w :** $w^{n+1} \leftarrow \sum_{j \leq 0} (-a_j E^{-j} u^{n+j} + \Delta T c_j E^{-j} N(u^{n+j}))$ $\triangleright$ (3.2), (3.1)
- 2: **function** $\Lambda(u_0, \dots, u_{S-1}; N_0, \dots, N_{S-1})$ $\triangleright$ reconstruction on a stencil, newest level first
- 3: $\hat{u}^{(m)} \leftarrow \Delta T^{-m} \sum_i W_{mi} u_i$, $N_{\text{FD}}^{(m)} \leftarrow \Delta T^{-m} \sum_i W_{mi} N_i$ for $m = 1, \dots, S-1$;
 $\hat{u}^{(S)} \leftarrow L \hat{u}^{(S-1)} + N_{\text{FD}}^{(S-1)}$ $\triangleright$ (4.3), (4.10)
- 4: $\hat{N}^{(S)} \leftarrow \sum_{j=0}^{S-1} \binom{S-1}{j} \hat{J}^{(j)} \hat{u}^{(S-j)}$; $\hat{N}^{(0)} \leftarrow N_0$; $\hat{N}^{(m)} \leftarrow N_{\text{FD}}^{(m)} - K_m \Delta T^{S-m} \hat{N}^{(S)}$,
 $m = 1, \dots, q-1$ $\triangleright$ (4.11), (4.18)
- 5: **return** $\hat{N}^{(m)}$, $m = 0, \dots, q-1$
- 6: **end function**
- 7: **Explicit correction:** $\hat{\hat{N}}^{(m)} \leftarrow \sum_{j \leq m} \binom{m}{j} (-L)^j \hat{N}^{(m-j)}$ from
 $\Lambda(u^n, \dots; N(u^n), \dots)$; $\hat{\delta} \leftarrow \sum_p \Delta T^p \gamma_p \hat{\hat{N}}^{(p-1)}$; $\tilde{u}^{n+1} \leftarrow E[w^{n+1} + \hat{\delta}]$ $\triangleright$ (4.22), predictor
- 8: **Shifted stencil:** $\hat{\hat{N}}_{n+1}^{(m)} \leftarrow \sum_{j \leq m} \binom{m}{j} (-L)^j \hat{\hat{N}}_{n+1}^{(m-j)}$ from
 $\Lambda(\tilde{u}^{n+1}, u^n, \dots; N(\tilde{u}^{n+1}), N(u^n), \dots)$ $\triangleright$ (4.35)
- 9: **Remainder:** $\hat{\rho} \leftarrow \sum_p \Delta T^p \gamma_p^+ \hat{\hat{N}}_{n+1}^{(p-1)} - \mathcal{J} \tilde{u}^{n+1}$ $\triangleright$ (3.35)
- 10: **Solve:** $\hat{u}^{n+1} \leftarrow [I - (1 - \theta)\mathcal{J}]^{-1} (E[w^{n+1} + \theta \hat{\delta}] + (1 - \theta)\hat{\rho})$ $\triangleright$ (3.36)
- 11: **Shift history:** push $\hat{u}^{n+1}$ and $N(\hat{u}^{n+1})$, drop level $S-1$

TABLE 5.1
Numerical testbeds and discretization parameters.

Testbed	Model	Discretization and parameters
Burgers [52]	$u_t + uu_x = \nu u_{xx}$	Periodic Fourier collocation; 65 modes; $\nu = 10^{-3}$; $\bar{u} = \frac{1}{2}$; $\max_k z_L = 0.03$ at the working step
Korteweg-de Vries [53]	$u_t + uu_x + \beta u_{xxx} = 0$	Periodic Fourier collocation; 65 modes; $\beta = 10^{-2}$; $\bar{u} = 1$; $\max_k z_L = 10$ at the working step
Lorenz-63 [15]	Linear part L and quadratic remainder N	$\sigma = 10$; $\rho = 28$; $b = \frac{8}{3}$; frozen operator $N'(u^n)$; flow-relative criterion (3.26) along the attractor

though stable, exhibits substantial phase error, as shown in Figure 5.4. Because $|e^{z_L}| = 1$, the integrating-factor scheme retains its $z_L = 0$ region, whereas the exact θ -correction produces the unbounded region, with bounded instability pockets, established in Proposition 3.2. The learned correction retains smaller regions whose boundaries are determined by the parasitic roots analyzed in Section 4.4.

TABLE 5.2
Common configuration of the numerical experiments.

Component	Configuration
Base schemes	IF-Euler; IF-AB2 with $\gamma = \frac{1}{2}$; IF-SBDF2 with $\gamma = 1$; IF-ARW $_{3/2}$ with $\gamma = \frac{3}{2}$
Correction	$q = S = 4$ and $\theta = \frac{1}{4}$
Compared variants	Explicit LMM; IMEX method [4]; integrating-factor method (3.3); exact θ -correction; learned θ -correction (3.36)
Working step	$\Delta T_{\text{scheme}} = \frac{1}{2} \Delta T_{\text{adm}}$
Admissibility criterion	$ \gamma_{q+1} z_{N,0}^{q+1} \leq 10^{-3}$ and $ \gamma_{q+2} z_{N,0}^{q+2} \leq \frac{1}{2} \gamma_{q+1} z_{N,0}^{q+1} $ for all modes
Reference solution	Exact- L fourth-order reference method with step $\Delta T/50$; the same solution initializes the multistep histories

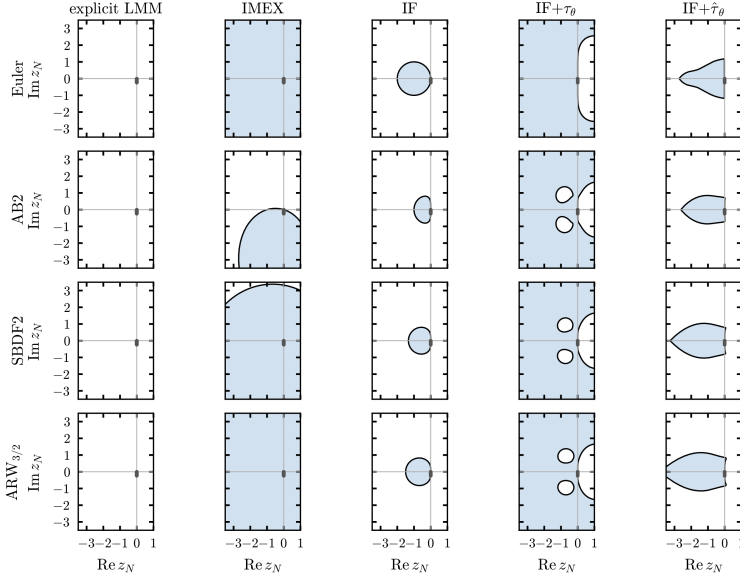


FIG. 5.1. Stability regions in the $z_{N,0}$ plane for the KdV testbed at the mode of largest $|z_L|$ ($z_L = 10i$): shaded $\max_r |g_r| \leq 1 + 10^{-3}$. Rows: base schemes; columns: explicit LMM, IMEX, IF, IF with the exact and with the learned θ -correction ($q = S = 4$, $\theta = \frac{1}{4}$). Dots: the modes $z_{N,0}(k)$ at $\Delta T_{\text{scheme}} = 3.8 \times 10^{-3}$.

The supplementary materials compare the raw and transported surrogate stencils and provide the corresponding grids for $z_L = 0$ and the Burgers testbed. Under the flow-relative criterion (3.26), Figure 5.2 shows that the frozen-Jacobian eigenvalues at the working step remain inside the regions of both θ -corrected variants for every base scheme.

Figure 5.3 shows the largest excess amplification over all modes on the Burgers testbed. Because z_L is small, the three base variants nearly coincide and lose stability before the admissible step, whereas the exact θ -correction remains stable throughout the displayed range and the learned θ -correction remains stable beyond the admissible

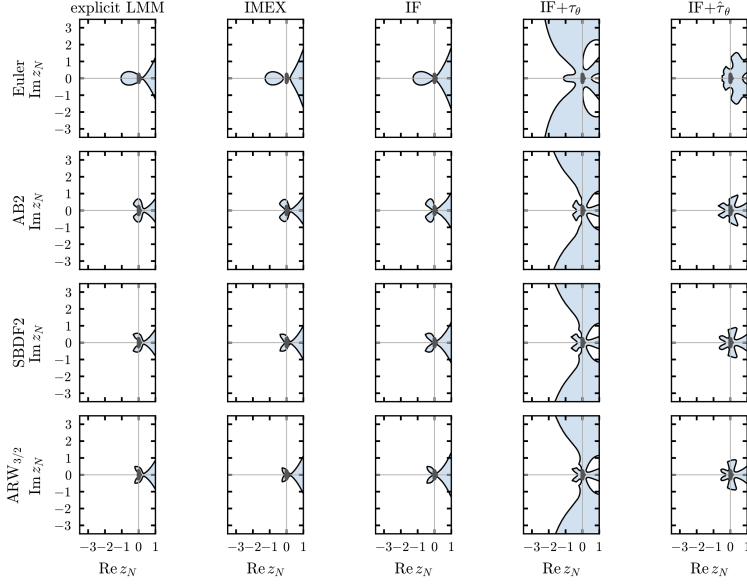


FIG. 5.2. *Stability regions for the Lorenz-63 system at the frozen-Jacobian mode of largest $|z_L|$ with the flow-relative criterion $\max_r |g_r| \leq |e^{z_L + z_N, 0}|(1 + 10^{-3})$; layout as in Figure 5.1. Dots: the frozen-Jacobian eigenvalues along the attractor at ΔT_{scheme} .*

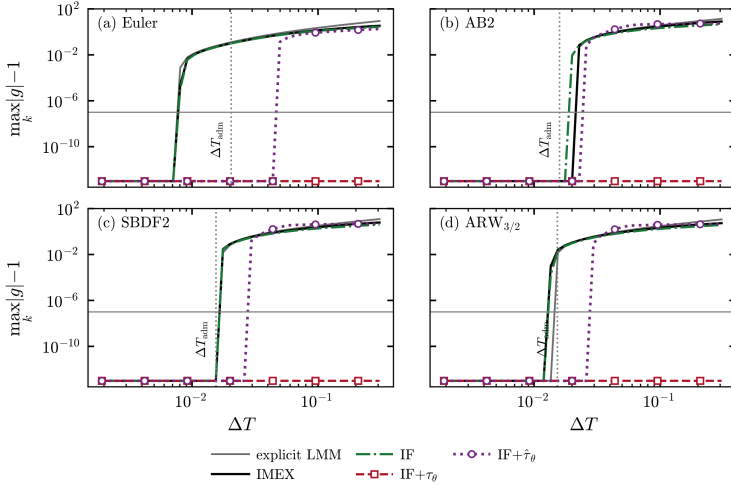


FIG. 5.3. *Burgers testbed: largest excess amplification $\max_k |g| - 1$ over the modes versus ΔT for the five variants with $q = S = 4$ and $\theta = \frac{1}{4}$; values below 10^{-13} are drawn at 10^{-13} ; horizontal line: tolerance 10^{-7} ; vertical line: ΔT_{adm} of the scheme.*

step for every base scheme. Consequently, accuracy, rather than stability, limits the practical step size of the corrected schemes.

Figure 5.4 confirms the predicted orders on the dispersive testbed: the integrating-factor scheme converges at order p_* , while both corrected variants converge at order four. The learned variant has a larger error constant because the inverse binomial map amplifies the finite-difference leak through powers of z_L , whereas the IMEX

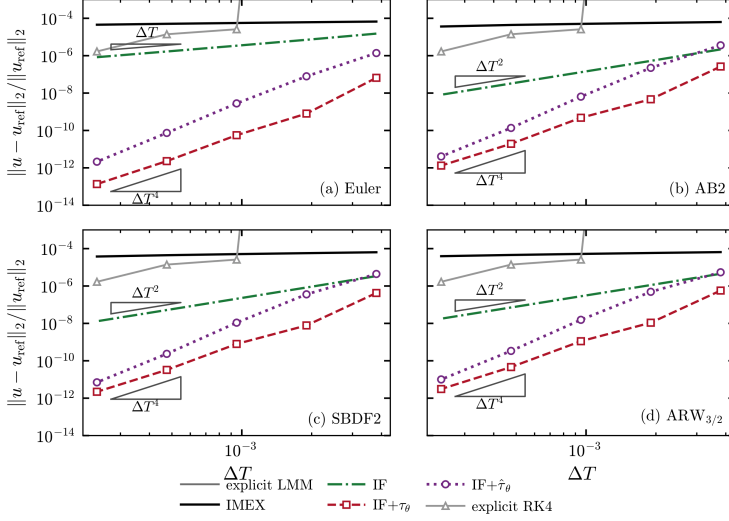


FIG. 5.4. *KdV testbed*: $\|u - u_{\text{ref}}\|_2 / \|u_{\text{ref}}\|_2$ at $t = 16 \Delta T_{\text{scheme}}$ versus ΔT for the five variants and explicit RK4.

error stagnates because the implicit component introduces phase error at large $|z_L|$. The explicit fourth-order reference method, in turn, becomes unstable as the step increases.

For Lorenz-63, Figure 5.5 shows that the learned and exact corrections are nearly indistinguishable at order four, and the supplementary Burgers results exhibit the same behavior. In the latter test, the gap between the learned and exact curves decays as ΔT^6 , consistently with (4.38).

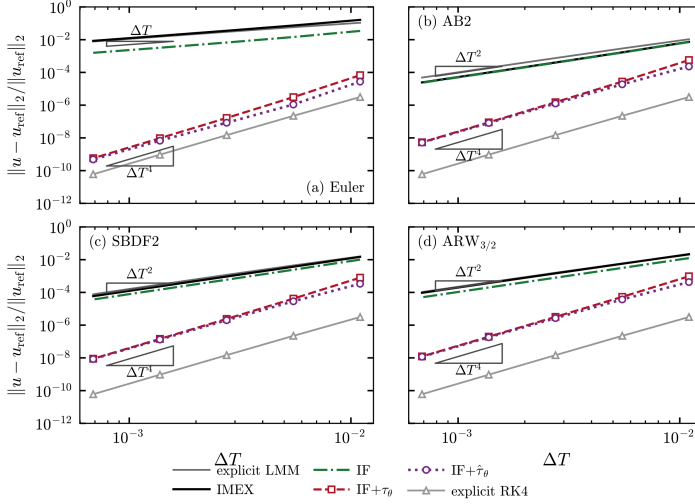


FIG. 5.5. *Lorenz-63 system*: $\|u - u_{\text{ref}}\|_2 / \|u_{\text{ref}}\|_2$ at $t = 16 \Delta T_{\text{scheme}}$ versus ΔT for the five variants and explicit RK4.

Figure 5.6 presents the Burgers error histories at the admissible step, which ex-

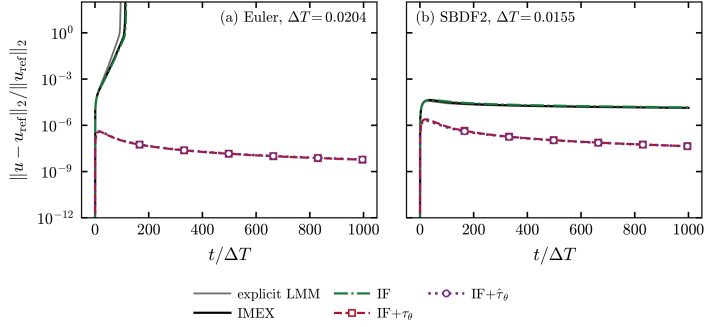


FIG. 5.6. *Burgers testbed*: $\|u - u_{\text{ref}}\|_2 / \|u_{\text{ref}}\|_2$ over 10^3 steps at $\Delta T = \Delta T_{\text{adm}}$ of each scheme for the five variants.

ceeds the stability limit of both the base schemes and the explicit learned correction. Accordingly, the base variants either diverge or retain a substantially larger error, whereas both θ -corrected variants remain accurate throughout the integration.

For the undamped KdV problem, the supplementary results show that the learned correction remains stable for IF-Euler but not for the two-step schemes, whose parasitic roots receive no linear damping. Figure 5.7 illustrates this limitation: under the explicit correction, the nonprincipal roots grow as $|z_{N,0}|^{1/3}$, in agreement with (4.33), while the implicit weight moves them toward the unit disk.

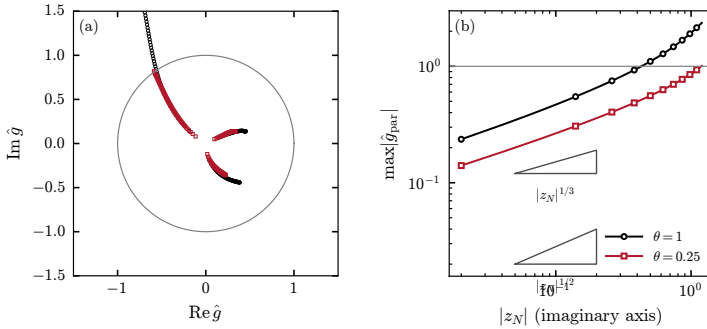


FIG. 5.7. *Parasitic roots of the learned IF-Euler scheme with $q = S = 4$, linear N and $z_L = 0$ along the imaginary $z_{N,0}$ axis.* (a) Non-principal roots for $\theta = 1$ and $\theta = \frac{1}{4}$. (b) Their largest modulus versus $|z_{N,0}|$ with the reference slopes $|z_{N,0}|^{1/3}$ and $|z_{N,0}|^{1/2}$.

Figures 5.8 and 5.9 compare accuracy as a function of the number of nonlinear evaluations, including every bilinear product in the exact rung ladders. On the dispersive testbed, the explicit fourth-order reference method is restricted by $|z_L|$, while ETDK4 [7, 8], although exact in L , has an error that depends on z_L and exhibits order reduction [14]. By contrast, the integrating-factor error is independent of z_L by Lemma 3.1, and the learned correction reaches a prescribed accuracy with substantially fewer nonlinear evaluations than the exact-rung variant, whose cost is dominated by construction of the derivative ladder.

For Lorenz-63, the learned IF-Euler correction has a cost-accuracy profile comparable to the fourth-order references. The supplementary materials show the corresponding comparison for Burgers.

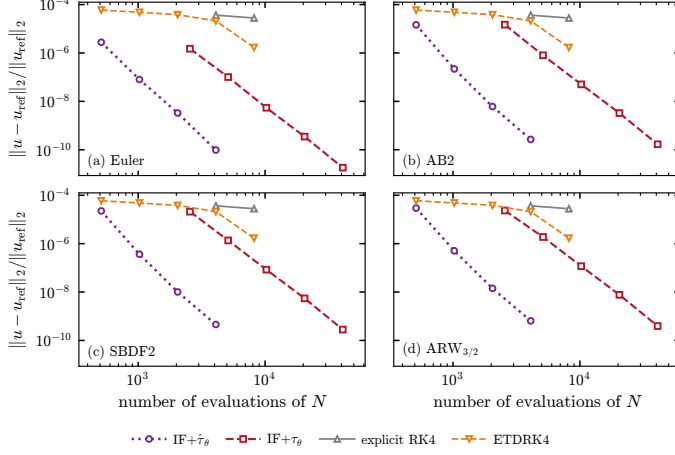


FIG. 5.8. *KdV testbed: $\|u - u_{\text{ref}}\|_2 / \|u_{\text{ref}}\|_2$ at $t = 256 \Delta T_{\text{scheme}}$ versus the total number of evaluations of N (bilinear products for the quadratic nonlinearity) needed to reach that time, for $\Delta T = \Delta T_{\text{scheme}} \times 2^{-j}$, $j = -1, \dots, 3$; one panel per base scheme, with explicit RK4 and ETD RK4 [7, 8] repeated in each panel. Per step: learned correction 2 (the surrogate action is not an evaluation of N), exact rungs 20 (two ladders of nine products plus two evaluations), Runge–Kutta methods 4.*

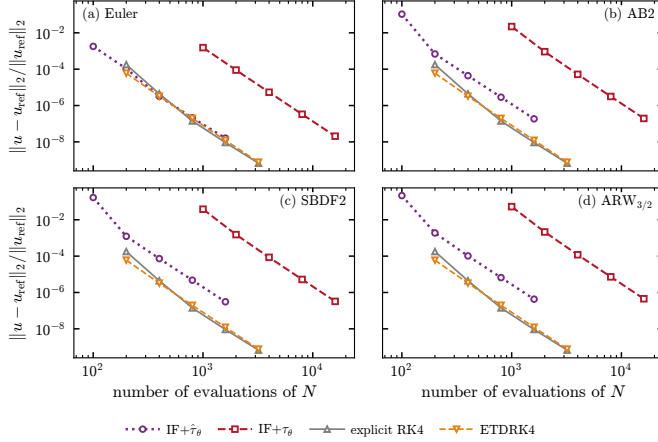


FIG. 5.9. *Lorenz-63 testbed: $\|u - u_{\text{ref}}\|_2 / \|u_{\text{ref}}\|_2$ at $t = 100 \Delta T_{\text{scheme}}$ versus the total number of evaluations of N , with the same counting and layout as Figure 5.8.*

The supplementary materials tabulate the modewise stability limits derived in Sections 3.5 and 4.4, reporting the largest admissible $|z_{N,0}|$ on the imaginary and negative real axes under the stated root tolerance. In both cases, the implicitly weighted learned correction enlarges the limit relative to the explicit learned correction.

6. Conclusion. We have developed a high-order correction framework for exponential linear multistep discretizations of semilinear evolution equations. By expressing each local-error term as a scalar multiple of a transformed nonlinear time derivative evaluated at either the current or the new time level, the moment representation yields an additive correction of any prescribed order q and separates the multistep error from the reconstruction error.

An implicit weight incorporates the new-level contribution through a frozen diagonal operator and thereby produces a rational modewise amplification factor. With exact rungs, the corresponding stability region is unbounded under the conditions of Proposition 3.2; with reconstructed rungs, it remains larger than that of the explicit correction, at the additional cost of one nonlinear evaluation and one division per mode.

The reconstruction combines stored state and nonlinear histories with a surrogate for the highest unresolved derivative. The analysis establishes the history-depth condition $S \geq q$ and identifies the leading surrogate-induced error through the second finite-difference moment, while the companion polynomial quantifies the perturbation of the principal amplification factor and provides a modewise admissibility condition for the surrogate error. The same polynomial also reveals the parasitic roots introduced by the stored history and the weights that control them.

For quadratic nonlinearities, these results reduce to explicit polynomial expressions, while for translation-equivariant polynomial and linear convolution surrogates, the supplementary materials provide both the approximation error and the Wiener-optimal irreducible error.

The Burgers, KdV, and Lorenz-63 experiments confirmed the predicted orders, enlarged stability regions, and reduced computational cost. The principal open questions concern the anti-dissipative reconstruction leak in undamped problems, the interaction between history depth and surrogate capacity, and the extension of the framework to third-order base schemes.

REFERENCES

- [1] G. Dahlquist, Convergence and stability in the numerical integration of ordinary differential equations, *Math. Scand.* **4** (1956), 33–53.
- [2] E. Hairer, S. P. Nørsett, G. Wanner, *Solving Ordinary Differential Equations I*, 2nd ed., Springer, 1993.
- [3] J. D. Lambert, *Numerical Methods for Ordinary Differential Systems*, Wiley, 1991.
- [4] U. M. Ascher, S. J. Ruuth, B. T. R. Wetton, Implicit–explicit methods for time-dependent partial differential equations, *SIAM J. Numer. Anal.* **32**(3) (1995), 797–823.
- [5] J. D. Lawson, Generalized Runge–Kutta processes for stable systems with large Lipschitz constants, *SIAM J. Numer. Anal.* **4** (1967), 372–380.
- [6] M. Hochbruck, A. Ostermann, Exponential integrators, *Acta Numer.* **19** (2010), 209–286.
- [7] S. M. Cox, P. C. Matthews, Exponential time differencing for stiff systems, *J. Comput. Phys.* **176** (2002), 430–455.
- [8] A.-K. Kassam, L. N. Trefethen, Fourth-order time-stepping for stiff PDEs, *SIAM J. Sci. Comput.* **26** (2005), 1214–1233.
- [9] E. Hairer, G. Wanner, *Solving Ordinary Differential Equations II: Stiff and Differential-Algebraic Problems*, 2nd ed., Springer, 1996.
- [10] D. W. Peaceman, H. H. Rachford, The numerical solution of parabolic and elliptic differential equations, *J. Soc. Indust. Appl. Math.* **3** (1955), 28–41.
- [11] J. Crank, P. Nicolson, A practical method for numerical evaluation of solutions of partial differential equations of the heat-conduction type, *Proc. Cambridge Philos. Soc.* **43** (1947), 50–67.
- [12] G. Dahlquist, A special stability problem for linear multistep methods, *BIT* **3** (1963), 27–43.
- [13] H. H. Rosenbrock, Some general implicit processes for the numerical solution of differential

- equations, *Comput. J.* **5** (1963), 329–330.
- [14] M. Hochbruck, A. Ostermann, Explicit exponential Runge–Kutta methods for semilinear parabolic problems, *SIAM J. Numer. Anal.* **43** (2005), 1069–1090.
 - [15] E. N. Lorenz, Deterministic nonperiodic flow, *J. Atmos. Sci.* **20** (1963), 130–141.
 - [16] C. Canuto, M. Y. Hussaini, A. Quarteroni, T. A. Zang, *Spectral Methods: Fundamentals in Single Domains*, Springer, 2006.
 - [17] G. M. Constantine, T. H. Savits, A multivariate Faà di Bruno formula with applications, *Trans. Amer. Math. Soc.* **348** (1996), 503–520.
 - [18] W. P. Johnson, The curious history of Faà di Bruno’s formula, *Amer. Math. Monthly* **109** (2002), 217–234.
 - [19] A. Griewank, A. Walther, *Evaluating Derivatives*, 2nd ed., SIAM, 2008.
 - [20] J. Kim, P. Moin, Application of a fractional-step method to incompressible Navier–Stokes equations, *J. Comput. Phys.* **59** (1985), 308–323.
 - [21] D. R. Durran, The third-order Adams–Bashforth method: an attractive alternative to leapfrog time differencing, *Mon. Weather Rev.* **119** (1991), 702–720.
 - [22] J. Frank, W. Hundsdorfer, J. G. Verwer, On the stability of implicit–explicit linear multistep methods, *Appl. Numer. Math.* **25** (1997), 193–205.
 - [23] W. Hundsdorfer, S. J. Ruuth, IMEX extensions of linear multistep methods with general monotonicity and boundedness properties, *J. Comput. Phys.* **225** (2007), 2016–2042.
 - [24] U. M. Ascher, S. J. Ruuth, R. J. Spiteri, Implicit–explicit Runge–Kutta methods for time-dependent partial differential equations, *Appl. Numer. Math.* **25** (1997), 151–167.
 - [25] C. A. Kennedy, M. H. Carpenter, Additive Runge–Kutta schemes for convection–diffusion–reaction equations, *Appl. Numer. Math.* **44** (2003), 139–181.
 - [26] L. Pareschi, G. Russo, Implicit–explicit Runge–Kutta schemes and applications to hyperbolic systems with relaxation, *J. Sci. Comput.* **25** (2005), 129–155.
 - [27] S. Boscarino, Error analysis of IMEX Runge–Kutta methods derived from differential-algebraic systems, *SIAM J. Numer. Anal.* **45** (2007), 1600–1621.
 - [28] J. C. Butcher, *Numerical Methods for Ordinary Differential Equations*, 3rd ed., Wiley, 2016.
 - [29] F. Bashforth, J. C. Adams, *An Attempt to Test the Theories of Capillary Action*, Cambridge Univ. Press, 1883.
 - [30] F. R. Moulton, *New Methods in Exterior Ballistics*, Univ. of Chicago Press, 1926.
 - [31] C. W. Gear, *Numerical Initial Value Problems in Ordinary Differential Equations*, Prentice-Hall, 1971.
 - [32] B. V. Minchev, W. M. Wright, A review of exponential integrators for first order semi-linear problems, Tech. Report 2/05, Department of Mathematics, NTNU, Trondheim, 2005.
 - [33] J. Certaine, The solution of ordinary differential equations with large time constants, in *Mathematical Methods for Digital Computers*, A. Ralston, H. S. Wilf, eds., Wiley, 1960, pp. 128–132.
 - [34] S. P. Nørsett, An A-stable modification of the Adams–Bashforth methods, in *Conference on the Numerical Solution of Differential Equations*, Lecture Notes in Math. 109, Springer, 1969, pp. 214–219.
 - [35] M. P. Calvo, C. Palencia, A class of explicit multistep exponential integrators for semilinear problems, *Numer. Math.* **102** (2006), 367–381.
 - [36] M. Hochbruck, A. Ostermann, Exponential multistep methods of Adams-type, *BIT* **51** (2011), 889–908.
 - [37] S. Krogstad, Generalized integrating factor methods for stiff PDEs, *J. Comput. Phys.* **203** (2005), 72–88.
 - [38] G. Beylkin, J. M. Keiser, L. Vozovoi, A new class of time discretization schemes for the solution of nonlinear PDEs, *J. Comput. Phys.* **147** (1998), 362–387.
 - [39] M. Hochbruck, C. Lubich, H. Selhofer, Exponential integrators for large systems of differential equations, *SIAM J. Sci. Comput.* **19** (1998), 1552–1574.
 - [40] M. Tokman, Efficient integration of large stiff systems of ODEs with exponential propagation iterative (EPI) methods, *J. Comput. Phys.* **213** (2006), 748–776.
 - [41] Y. Bar-Sinai, S. Hoyer, J. Hickey, M. P. Brenner, Learning data-driven discretizations for partial differential equations, *Proc. Natl. Acad. Sci. USA* **116** (2019), 15344–15349.
 - [42] D. Kochkov, J. A. Smith, A. Alieva, Q. Wang, M. P. Brenner, S. Hoyer, Machine learning–accelerated computational fluid dynamics, *Proc. Natl. Acad. Sci. USA* **118** (2021), e2101784118.
 - [43] K. Um, R. Brand, Y. Fei, P. Holl, N. Thuerey, Solver-in-the-loop: learning from differentiable physics to interact with iterative PDE-solvers, in *Advances in Neural Information Processing Systems* 33, 2020.
 - [44] B. List, L.-W. Chen, N. Thuerey, Learned turbulence modelling with differentiable fluid solvers:

- physics-based loss functions and optimisation horizons, *J. Fluid Mech.* **949** (2022), A25.
- [45] A. Gupta, P. F. J. Lermusiaux, Neural closure models for dynamical systems, *Proc. R. Soc. A* **477** (2021), 20201004.
 - [46] K. Duraisamy, G. Iaccarino, H. Xiao, Turbulence modeling in the age of data, *Annu. Rev. Fluid Mech.* **51** (2019), 357–377.
 - [47] S. L. Brunton, B. R. Noack, P. Koumoutsakos, Machine learning for fluid mechanics, *Annu. Rev. Fluid Mech.* **52** (2020), 477–508.
 - [48] R. T. Q. Chen, Y. Rubanova, J. Bettencourt, D. K. Duvenaud, Neural ordinary differential equations, in *Advances in Neural Information Processing Systems 31*, 2018.
 - [49] M. Schetzen, *The Volterra and Wiener Theories of Nonlinear Systems*, Wiley, 1980.
 - [50] N. Wiener, *Extrapolation, Interpolation, and Smoothing of Stationary Time Series*, MIT Press, 1949.
 - [51] J. Douglas, H. H. Rachford, On the numerical solution of heat conduction problems in two and three space variables, *Trans. Amer. Math. Soc.* **82** (1956), 421–439.
 - [52] J. M. Burgers, A mathematical model illustrating the theory of turbulence, *Adv. Appl. Mech.* **1** (1948), 171–199.
 - [53] D. J. Korteweg, G. de Vries, On the change of form of long waves advancing in a rectangular canal, and on a new type of long stationary waves, *Philos. Mag.* **39** (1895), 422–443.
 - [54] S. A. Orszag, On the elimination of aliasing in finite-difference schemes by filtering high-wavenumber components, *J. Atmos. Sci.* **28** (1971), 1074.
 - [55] L. N. Trefethen, *Spectral Methods in MATLAB*, SIAM, 2000.
 - [56] J. G. Charney, R. Fjørtoft, J. von Neumann, Numerical integration of the barotropic vorticity equation, *Tellus* **2** (1950), 237–254.
 - [57] W. E. Milne, *Numerical Calculus*, Princeton Univ. Press, 1949.